# A complete solution to the Grünbaum–Loewner centroid problems

**Shouda Wang**[1] **Ge Xiong**[2] **Kaiwen Yang**[2]

[1]Program in Applied and Computational Mathematics, Princeton University, Princeton, NJ 08544, USA
[2]School of Mathematical Sciences, Tongji University, Shanghai 200092, P. R. China

**Abstract:** In the 1960s, B. Grünbaum and C. Loewner raised problems concerning the minimum possible number of hyperplane sections of a convex body in $\mathbb{R}^n$ whose centroids are required to coincide with the centroid of the body itself. By employing the hemispherical transform and Morse theory, we provide a complete solution to these problems.

**2020 Mathematics Subject Classification:** 52A20, 33C55, 57R70.

**Keywords:** Convex body, hyperplane section, centroid, hemispherical transform, spherical harmonics, Morse function.

## 1. Introduction

Let $K$ be a convex body (i.e., a compact convex set with non-empty interior) in the $n-$dimensional Euclidean space, $\mathbb{R}^n$. The *centroid* (also called *center of mass*, or *barycenter*) of $K$ is the point

$$c(K) = \frac{1}{\mathcal{H}^n(K)} \int_K x \, d\mathcal{H}^n(x),$$

where $\mathcal{H}^n$ denotes the $n$-dimensional Hausdorff measure. A *hyperplane section* of $K$ is a set $K \cap H$, where $H$ is a hyperplane in $\mathbb{R}^n$ and $\mathcal{H}^{n-1}(K \cap H) > 0$. So, the centroid of $K \cap H$ is

$$c(K \cap H) = \frac{1}{\mathcal{H}^{n-1}(K \cap H)} \int_{K \cap H} x \, d\mathcal{H}^{n-1}(x).$$

A classical result using topology asserts that every planar convex body $K \subseteq \mathbb{R}^2$ has at least *three* hyperplane sections whose centroids coincide with the centroid $c(K)$ of $K$. Please refer to Bose [2] and Ehrhart [4]. This led Grünbaum to ask whether the same phenomenon persists in arbitrary dimension [9, p. 41] in 1961.

**Problem 1** (Grünbaum)**.** *Let $K$ be a convex body in $\mathbb{R}^n$. Do there exist at least $n+1$ hyperplane sections whose centroids coincide with $c(K)$?*

In 1965, Loewner asked for the largest integer that can replace $n+1$ in the Grünbaum problem [5, Problem 28].

**Problem 2** (Loewner)**.** *For each integer $n \geq 2$, define $\mu(n)$ as the largest integer with the property that every convex body $K \subseteq \mathbb{R}^n$ admits at least $\mu(n)$ hyperplane sections whose centroids coincide with $c(K)$ of $K$. What is the value of $\mu(n)$?*

E-mail addresses: 1. shoudawang@princeton.edu; 2. xiongge@tongji.edu.cn; 3. yangkaiwen@tongji.edu.cn.

Hence, the Grünbaum problem asks whether $\mu(n) \geq n+1$ always holds and the classical topological result is interpreted as $\mu(2) \geq 3$. By contrast, topological arguments only lead to a lower bound $\mu(n) \geq 1$ for dimensions $n \geq 3$. Please refer to Section 2 for its explanations.

The discrepancy between the known lower bound $\mu(n) \geq 1$ and the Grünbaum conjectured lower bound $\mu(n) \geq n+1$ persisted for a long time, until Myroshnychenko, Tatarko and Yaskin [17] recently *disproved* the Grünbaum conjecture in dimensions $n \geq 5$. Precisely, they [17] proved that $\mu(n) = 1$ for all $n \geq 5$ by using non-intersection bodies. Since non-intersection bodies do not exist in dimensions 3 and 4 (see Gardner [7] and Zhang [23]), their approach cannot cover dimensions 3 and 4. Despite this, they conjectured that $\mu(3) = \mu(4) = 1$ as well [17, Remark 1].

It is interesting that Huang, Myroshnychenko, Tatarko and Yaskin [13] very recently studied a hyperbolic analogue $\mu_{\mathrm{Hyp}}$ of $\mu$, and proved that $\mu_{\mathrm{Hyp}}(2) = 3$ and $\mu_{\mathrm{Hyp}}(n) = 1$ for all $n \geq 3$ by using the existence of hyperbolic analogues of non-intersection bodies established by Yaskin [22].

In this article, we prove that $\mu(n) \leq 1$ for all dimensions $n \geq 3$. This, together with established results, provides a *complete* solution to the Grünbaum–Loewner problems.

**Theorem 1.1.** *For each $n \geq 3$, there exists a smooth strictly convex body $K$ in $\mathbb{R}^n$ for which exactly one hyperplane section has its centroid coinciding with the centroid $c(K)$ of $K$. Therefore,*

$$\mu(n) = \begin{cases} 3, & n = 2, \\ 1, & n \geq 3. \end{cases}$$

Recall that a convex body $K$ in $\mathbb{R}^n$ is called *smooth* if $\partial K$ is a $C^\infty$ submanifold of $\mathbb{R}^n$; it is called *strictly convex* if for all $x, y \in K$ and every $\lambda \in (0,1)$, one has $\lambda x + (1-\lambda)y \in \operatorname{int} K$.

We now briefly outline the key ideas underlying the proof of the main results.

First, for a convex body $K$ in $\mathbb{R}^n$, its *half-space volume function* is $A_K : \mathbb{S}^{n-1} \to \mathbb{R}$,

$$A_K(u) = \mathcal{H}^n\big(K \cap \{x \in \mathbb{R}^n : \langle x - c(K), u\rangle \geq 0\}\big), \quad u \in \mathbb{S}^{n-1},$$

where $\mathbb{S}^{n-1}$ is the unit sphere of $\mathbb{R}^n$. Here and throughout this article, $\langle \cdot, \cdot \rangle$ denotes the standard inner product in $\mathbb{R}^n$. Suppose the centroid $c(K)$ of $K$ is at the origin $o$. Then $A_K$ has the property that its critical points are precisely the normal vectors to hyperplane sections whose centroids lie at the origin $o$ (Lemma 2.1). Thus, to prove Theorem 1.1, it suffices to exhibit a convex body $K$ for which $A_K$ possesses exactly two antipodal critical points.

Second, we invoke the *hemispherical transform* $\mathrm{T}$, which links the half-space volume function $A_K$ to the *radial function* $\rho_K$ of $K$ by

$$A_K = \frac{1}{n}\,\mathrm{T}(\rho_K^n),$$

where $\rho_K(u) = \sup\{\lambda > 0 : \lambda u \in K\}$, $u \in \mathbb{S}^{n-1}$. We prove that $\mathrm{T}$ is an *isomorphism* on the space of smooth odd functions on $\mathbb{S}^{n-1}$, and that $\mathrm{T} f$ lacks a *degree-one* spherical-harmonic component if and only if $f$ does (Lemma 3.2). So, if we can find a smooth odd Morse function $f$ on $\mathbb{S}^{n-1}$ with exactly two critical points and no degree-one component, then solving $\mathrm{T}\, h = f$ gives an $h$ with the *same* property.

Third, for small $\varepsilon \in \mathbb{R}$ define

$$K_\varepsilon = \{ru : \; 0 \le r \le (1+\varepsilon h(u))^{1/(n+1)}, \; u \in \mathbb{S}^{n-1}\}.$$

The absence of the degree-one component in $h$ ensures that $c(K_\varepsilon) = o$, while the expansion of $A_{K_\varepsilon}$ shows that, after rescaling and subtracting a constant, $A_{K_\varepsilon} \to f$ in $C^2$ as $\varepsilon \to 0$. The stability of Morse functions (Lemma 4.3) then implies that $A_{K_\varepsilon}$ has the same number of critical points as $f$, namely two. Consequently, $K_\varepsilon$ has exactly one hyperplane section whose centroid agrees with $c(K_\varepsilon)$, as desired.

Finally, it remains to construct such a Morse function $f$. This is accomplished by twisting a height function on the unit sphere $\mathbb{S}^{n-1}$; the details are given in Lemma 5.3.

*Remark* 1.2. Grünbaum [9] further posed a weaker question: does every convex body $K \subseteq \mathbb{R}^n$ contains a point $p \in \operatorname{int} K$ that serves as the centroid of at least $n+1$ hyperplane sections of $K$? Problem 1 then asks whether we may take $p = c(K)$ in this statement. Grünbaum claimed a proof of this weaker statement in [10, Sec. 6.2]; however, Patáková, Tancer and Wagner [18] found a substantive flaw in his arguments. Nonetheless, the same authors [18] showed that the weaker statement does hold in dimension $n = 3$. The cases $n \ge 4$ remain open to date.

**Acknowledgments.** The results in this paper are obtained independently and simultaneously by Shouda Wang (at Princeton University) and by Ge Xiong and Kaiwen Yang (at Tongji University in Shanghai). The first author thanks Vlad Yaskin for his comments on an earlier draft. The research of the last two authors was supported by NSFC No. 12271407.

## 2. Preliminaries

Let $K$ be a convex body in $\mathbb{R}^n$ with centroid $c(K)$. For $u \in \mathbb{S}^{n-1}$, write

$$u^\perp = \{x \in \mathbb{R}^n : \langle x, u\rangle = 0\} \quad \text{and} \quad u^\perp + c(K) = \{x + c(K) : x \in u^\perp\}.$$

Then $K \cap (u^\perp + c(K))$ is a hyperplane section passing through centroid $c(K)$. Concerning the Grünbaum–Loewner centroid problems, we naturally restrict on such hyperplane sections.

For a smooth convex body $K$ in $\mathbb{R}^n$, its radial function $\rho_K \in C^\infty(\mathbb{S}^{n-1})$. If $K$ is a smooth convex body with positive Gauss curvature everywhere, then $K$ is strictly convex.

### 2.1. The half-space volume function.

A crucial ingredient of our approach is the following formula (see, e.g., Haddad, Jiménez and Villa [12, Lemma 8], and Patáková, Tancer and Wagner [18, Prop. 11]) of the half-space volume function $A_K(u) = \mathcal{H}^n(K \cap \{x \in \mathbb{R}^n : \langle x - c(K), u\rangle \ge 0\})$.

**Lemma 2.1.** *If $K \subseteq \mathbb{R}^n$ is a convex body with $c(K) = o$ and $u \in \mathbb{S}^{n-1}$, then $A_K$ is of class $C^1$ and its directional derivative in $v \in T_u\mathbb{S}^{n-1} = u^\perp$ at $u$ is*

$$D_v A_K(u) = \int_{K \cap u^\perp} \langle x, v\rangle \, d\mathcal{H}^{n-1}(x).$$

*Consequently, $u$ is a critical point of $A_K$ if and only if $c(K \cap u^\perp) = c(K) = o$.*

Since $A_K(-u) = \mathcal{H}^n(K) - A_K(u)$ for all $u \in \mathbb{S}^{n-1}$, it follows that the critical points of $A_K$ come in antipodal pairs.

The well-known bound $\mu(n) \geq 1$ is an immediate consequence of Lemma 2.1 (see Grünbaum [9, Theorem 1], Meyer and Reisner [15, Lemma 8], or Steinhaus [21, (1)]).

**Lemma 2.2.** *If $n \geq 2$ and $K \subseteq \mathbb{R}^n$ is a convex body with centroid $c(K) = o$, then there exists a direction $u_0 \in \mathbb{S}^{n-1}$ such that $c(K \cap u_0^\perp) = o$. In other words, $\mu(n) \geq 1$.*

*Proof.* By Lemma 2.1, the function $A_K$ is of class $C^1$. Since $\mathbb{S}^{n-1}$ is compact and has no boundary, $A_K$ attains its maximum at some direction $u_0 \in \mathbb{S}^{n-1}$ and $dA_K(u_0) = o$. By Lemma 2.1 again, we have $c(K \cap u_0^\perp) = o = c(K)$. Equivalently, $\mu(n) \geq 1$. □

2.2. **Perturbative estimates.**

The following lemma states that any sufficiently small $C^2$ perturbation of the radial function of a smooth convex body with positive Gauss curvature everywhere is still a radial function of a smooth convex body with positive Gauss curvature everywhere.

**Lemma 2.3.** *If $\rho_0 \in C^\infty(\mathbb{S}^{n-1})$ is a radial function of a smooth convex body with positive Gauss curvature everywhere, then there exists $\varepsilon > 0$ such that for all $\rho \in C^\infty(\mathbb{S}^{n-1})$ with $\|\rho - \rho_0\|_{C^2(\mathbb{S}^{n-1})} < \varepsilon$, the set*

$$K_\rho = \{ru : u \in \mathbb{S}^{n-1},\ 0 \leq r \leq \rho(u)\}$$

*is also a smooth convex body with positive Gauss curvature everywhere.*

*Proof.* This is a standard perturbation argument, see e.g., Koldobsky [14, p. 96] or Myroshnychenko, Tatarko and Yaskin [17, Lemma 4].

Let $\sigma_{ij}$ be the standard metric on $\mathbb{S}^{n-1}$, and let $\nabla$ be its Levi–Civita connection. In local coordinates, the second fundamental form of $\partial K_\rho$ is

$$\mathrm{II}^\rho_{ij} = \frac{\rho^2\sigma_{ij} + 2\nabla_i\rho\,\nabla_j\rho - \rho\nabla_{ij}\rho}{\sqrt{\rho^2 + |\nabla\rho|^2}}.$$

See Guan and Spruck [11, §2, equations (2.3)].

For $\rho = \rho_0$, $\mathrm{II}^{\rho_0}$ is positive definite since $\rho_0$ is the radial function of a smooth convex body with positive Gauss curvature everywhere. Since $\mathbb{S}^{n-1}$ is compact and $\mathrm{II}^\rho$ depends continuously on $\rho$ in the $C^2$-topology, $\mathrm{II}^\rho$ is positive definite for all $\rho$ with $\|\rho - \rho_0\|_{C^2(\mathbb{S}^{n-1})} < \varepsilon$ for sufficiently small $\varepsilon > 0$. Therefore, $K_\rho$ is a smooth convex body with positive Gauss curvature everywhere. □

**Lemma 2.4.** *If $h \in C^\infty(\mathbb{S}^{n-1})$ and $\beta > 0$, then $\|(1+\varepsilon h)^\beta - 1 - \beta\varepsilon h\|_{C^2(\mathbb{S}^{n-1})} = O(\varepsilon^2)$, as $\varepsilon \to 0$.*

*Proof.* Let

$$\varphi(s) = (1+s)^\beta - 1 - \beta s, \quad s > -1.$$

Since $\varphi(0) = \varphi'(0) = 0$, there exists $\delta > 0$ and a smooth function $\psi \in C^\infty(-\delta, \delta)$ such that $\varphi(s) = s^2\psi(s)$ for $|s| < \delta$. Let $M_0 = \|h\|_{C^0(\mathbb{S}^{n-1})}$. If $M_0 = 0$, the claim is trivial. Otherwise, choose $\varepsilon_0 = \frac{\delta}{2M_0}$. So for all $|\varepsilon| \leq \varepsilon_0$, $|\varepsilon h(u)| \leq \frac{\delta}{2}$ for all $u \in \mathbb{S}^{n-1}$, and therefore the function

$$(1+\varepsilon h)^\beta - 1 - \beta\varepsilon h = \varepsilon^2 h^2 \psi(\varepsilon h)$$

is well-defined on $\mathbb{S}^{n-1}$.

In the following, we show that $\|h^2\psi(\varepsilon h)\|_{C^2(\mathbb{S}^{n-1})}$ is bounded uniformly for $|\varepsilon| \le \varepsilon_0$. Let

$$A_j = \sup_{|t|\le\delta/2} |\psi^{(j)}(t)|, \qquad j = 0, 1, 2.$$

Then $A_0, A_1, A_2 < \infty$ by $\psi \in C^\infty(-\delta, \delta)$. For brevity, write $F_\varepsilon = h^2\psi(\varepsilon h)$. Then

$$|F_\varepsilon| \le A_0 \|h\|_{C^0(\mathbb{S}^{n-1})} = A_0 M_0^2.$$

Since

$$\nabla F_\varepsilon = (2h\psi(\varepsilon h) + \varepsilon h^2\psi'(\varepsilon h))\nabla h,$$

it follows that $\|\nabla F_\varepsilon\|_{C^0(\mathbb{S}^{n-1})}$ is bounded uniformly for $|\varepsilon| \le \varepsilon_0$ by $h \in C^\infty(\mathbb{S}^{n-1})$ and $|\varepsilon h| \le \frac{\delta}{2}$.

Finally,

$$\nabla^2 F_\varepsilon = (2\psi(\varepsilon h) + 4\varepsilon h\psi'(\varepsilon h) + \varepsilon^2 h^2 \psi''(\varepsilon h))\nabla h \otimes \nabla h + (2h\psi(\varepsilon h) + \varepsilon h^2\psi'(\varepsilon h))\nabla^2 h,$$

it also follows that $\|\nabla^2 F_\varepsilon\|_{C^0(\mathbb{S}^{n-1})}$ is bounded uniformly for $|\varepsilon| \le \varepsilon_0$.

Hence, there exists $C > 0$ independent of $\varepsilon$, such that

$$\|F_\varepsilon\|_{C^2(\mathbb{S}^{n-1})} \le C, \quad \text{for all } |\varepsilon| \le \varepsilon_0.$$

Therefore,

$$\left\|(1+\varepsilon h)^\beta - 1 - \beta\varepsilon h\right\|_{C^2(\mathbb{S}^{n-1})} = |\varepsilon|^2 \|F_\varepsilon\|_{C^2(\mathbb{S}^{n-1})} \le C|\varepsilon|^2, \quad \text{for all } |\varepsilon| \le \varepsilon_0.$$

This proves the desired estimate. □

2.3. **The planar case.**

For sake of the completeness, we include the proof of the planer case, i.e., $\mu(2) = 3$, in this part. For the result $\mu(2) \ge 3$, one can refer to Bose [2], Ehrhart [4], and Myroshnychenko, Tatarko and Yaskin [17, Sec. 3]; For the result $\mu(2) \le 3$, we give a short proof.

**Lemma 2.5.** $\mu(2) \ge 3$.

*Proof.* Let $K \subseteq \mathbb{R}^2$ be a convex body. W.l.o.g., assume $c(K) = o$. Write $e_\alpha = (\cos\alpha, \sin\alpha)$ and

$$g(\alpha) = \rho_K(e_\alpha)^3 - \rho_K(e_{\alpha+\pi})^3, \quad \alpha \in \mathbb{R}.$$

Then $g(\alpha) = 0$ if and only if $c(K \cap e_{\alpha+\pi/2}^\perp) = o$. Since $c(K) = o$, by $g(s+\pi) = -g(s)$ and the polar coordinate formula of integral, for each $\alpha \in \mathbb{R}$ we have

$$\int_\alpha^{\alpha+\pi} g(s) e_s \, ds = \int_0^{2\pi} \rho_K(e_s)^3 e_s \, ds = 3\int_K x \, dx = o.$$

So,

$$\int_\alpha^{\alpha+\pi} g(s)\langle e_s, e_{\alpha+\pi/2}\rangle \, ds = \int_\alpha^{\alpha+\pi} g(s)\sin(s-\alpha)\, ds = 0.$$

Since $\sin(s-\alpha) > 0$ in $(\alpha, \alpha+\pi)$, it follows that $g$ is identically zero, or changes sign in $(\alpha, \alpha+\pi)$.

If $g$ is identically zero, then $c(K \cap e_{\alpha+\pi/2}^\perp) = o$ everywhere. Assume $g(s) \ne 0$ for some $s$.

Since $g(s+\pi) = -g(s)$, there exists $\theta_1 \in (s, s+\pi)$ with $g(\theta_1) = 0$. Since $g$ changes sign in $(\alpha, \alpha+\pi)$ for each $\alpha \in \mathbb{R}$, there exists $\theta_2 \in (\theta_1, \theta_1+\pi)$ with $g(\theta_2) = 0$. Assume $\theta_2$ is the only zero of $g$ in $(\theta_1, \theta_1+\pi)$, w.l.o.g., assume $g > 0$ in $(\theta_1, \theta_2)$ and $g < 0$ in $(\theta_2, \theta_1+\pi)$. By $g(s+\pi) = -g(s)$, it follows that $g < 0$ in $(\theta_1+\pi, \theta_2+\pi)$, and therefore $g \le 0$ in $(\theta_2, \theta_2+\pi)$, contradicting that $g$ changes sign in $(\alpha, \alpha+\pi)$ for each $\alpha \in \mathbb{R}$. So, there exists $\theta_3 \in (\theta_1, \theta_1+\pi) \setminus \{\theta_2\}$ with $g(\theta_3) = 0$.

Hence,

$$c(K \cap e_{\theta_1+\pi/2}^{\perp}) = c(K \cap e_{\theta_2+\pi/2}^{\perp}) = c(K \cap e_{\theta_3+\pi/2}^{\perp}) = o,$$

which implies that $\mu(2) \ge 3$. □

**Lemma 2.6.** $\mu(2) \le 3$.

*Proof.* Let $S \subseteq \mathbb{R}^2$ be the equilateral triangle given by

$$S = \bigcap_{j=0}^{2} \{x \in \mathbb{R}^2 : \langle x, v_j \rangle \le 1\},$$

where $v_j = \left(\cos \frac{2\pi j}{3}, \sin \frac{2\pi j}{3}\right)$, $j = 0, 1, 2$. Then its centroid $c(S) = o$, and its radial function

$$\rho_S(u) = \frac{1}{\max_j \langle u, v_j \rangle}, \quad \text{for all } u \in \mathbb{S}^1.$$

Note that the chord $S \cap \operatorname{span}\{u\}$ of $S$ has centroid at origin $o$ iff $\rho_S(u) = \rho_S(-u)$, and iff

$$\max_j \langle u, v_j \rangle = \max_j \langle -u, v_j \rangle = -\min_j \langle u, v_j \rangle.$$

Let $M = \max_j \langle u, v_j \rangle$ and $m = \min_j \langle u, v_j \rangle$. Then $M = -m$.

If $M = 0$, then all the $\langle u, v_j \rangle = 0$, which is impossible for $u \ne 0$. Thus $M > 0$, and $m < 0$. Since $v_0 + v_1 + v_2 = 0$, it follows that $\langle u, v_0 \rangle + \langle u, v_1 \rangle + \langle u, v_2 \rangle = 0$ for each $u \in \mathbb{S}^1$, and therefore $\langle u, v_j \rangle = 0$ for some $j \in \{0, 1, 2\}$. So, the direction $u$ is parallel to the side of $S$ corresponding to $v_j$. Since $S$ has exactly three side directions, there are at most *three* distinct directions $u$ for which this can happen. Therefore, $\mu(2) \le 3$. □

## 3. **The hemispherical transform**

In this part, we present some properties of the hemispherical transform $\mathrm{T}$, which was posed by Funk [6] on $\mathbb{S}^2$ in 1915 and generalized to arbitrary dimensions by Rubin [19] in 1999. We prove that $\mathrm{T}$ is an isomorphism between the spaces $C^\infty_{\mathrm{odd}}(\mathbb{S}^{n-1})$ of *smooth odd functions* on $\mathbb{S}^{n-1}$ and is a *bounded* operator on $C^k(\mathbb{S}^{n-1})$ for each non-negative integer $k$.

**Definition 3.1.** The *hemispherical transform* of a function $f \in C(\mathbb{S}^{n-1})$ is defined by

$$\mathrm{T} f(u) = \int_{\mathbb{S}^{n-1}} \mathbb{1}_{\{\langle \theta, u \rangle \ge 0\}} f(\theta)\, d\mathcal{H}^{n-1}(\theta), \quad \forall\, u \in \mathbb{S}^{n-1}.$$

Here $\mathbb{1}$ denotes the indicator function. Obviously that the hemispherical transform $\mathrm{T}$ is linear.

Let $K$ be a convex body in $\mathbb{R}^n$ with $c(K) = o$. Then, for $\forall\ u \in \mathbb{S}^{n-1}$,

$$\begin{aligned} A_K(u) &= \mathcal{H}^n\big(K \cap \{x \in \mathbb{R}^n : \langle x, u\rangle \geq 0\}\big) \\ &= \int_{\mathbb{S}^{n-1}} \int_0^{\rho_K(\theta)} \mathbb{1}_{\{\langle \theta, u\rangle \geq 0\}} r^{n-1}\, dr\, d\mathcal{H}^{n-1}(\theta) \\ &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} \mathbb{1}_{\{\langle \theta, u\rangle \geq 0\}} \rho_K(\theta)^n\, d\mathcal{H}^{n-1}(\theta). \end{aligned}$$

That is,

$$A_K(u) = \frac{1}{n}\,\mathrm{T}(\rho_K^n)(u), \quad \forall\ u \in \mathbb{S}^{n-1}. \tag{3.1}$$

### 3.1. **Spherical harmonics.**

For later use, we collect some basic facts on spherical harmonics. One can refer to Groemer [8, Section 3], Schneider [20, Appendix], and Dai and Xu [3] for more details.

A *spherical harmonic* of degree $m$ on $\mathbb{S}^{n-1}$ is the restriction to $\mathbb{S}^{n-1}$ of a homogeneous harmonic polynomial of degree $m$ on $\mathbb{R}^n$. If $Y_m$ is a spherical harmonic of degree $m$, then

$$Y_m(-u) = (-1)^m Y_m(u) \quad \text{and} \quad \Delta_{\mathbb{S}^{n-1}} Y_m = -m(m+n-2) Y_m.$$

For $n \geq 3$, The set $\mathscr{H}^m$ of spherical harmonics of degree $m$ is a finite-dimensional vector subspace of $C^\infty(\mathbb{S}^{n-1})$ (and hence of $L^2(\mathbb{S}^{n-1})$), with dimension

$$\dim \mathscr{H}^m = N(n,m) = \frac{(2m+n-2)\Gamma(n+m-2)}{\Gamma(m+1)\Gamma(n-1)}.$$

For $n = 2$, $\dim \mathscr{H}^0 = 1$ and $\dim \mathscr{H}^m = 2$ for $m \geq 1$.

Assume the spherical harmonics are real-valued. For each $m \geq 0$, we choose an orthonormal basis $\{Y_{m,k}\}_{k=1}^{N(n,m)}$ of $\mathscr{H}^m$ with respect to the $L^2$ inner product on $\mathbb{S}^{n-1}$ defined by

$$(f, g) := \int_{\mathbb{S}^{n-1}} f(u) g(u)\, d\mathcal{H}^{n-1}(u), \quad \forall\ f, g \in L^2(\mathbb{S}^{n-1}).$$

It is well-known that spherical harmonics of different degrees are orthogonal, i.e.,

$$(f, g) = 0, \quad \forall\ f \in \mathscr{H}^k,\ g \in \mathscr{H}^j,\ k \neq j,$$

and the union $\bigoplus_{m=0}^\infty \mathscr{H}^m$ is dense in $L^2(\mathbb{S}^{n-1})$. Consequently, the collection

$$\{Y_{m,k} : m = 0,\ 1,\ 2, \ldots,\ k = 1, \ldots, N(n,m)\}$$

forms a complete orthonormal system of $L^2(\mathbb{S}^{n-1})$.

Let $\pi_m : L^2(\mathbb{S}^{n-1}) \to \mathscr{H}^m$ denote the orthogonal projection onto $\mathscr{H}^m$. With the chosen orthonormal basis, for $\forall\ f \in L^2(\mathbb{S}^{n-1})$, it is explicitly given by

$$\pi_m f = \sum_{k=1}^{N(n,m)} (f, Y_{m,k}) Y_{m,k} \quad \text{and} \quad f = \sum_{m=0}^\infty \pi_m f \quad \text{in } L^2(\mathbb{S}^{n-1}).$$

Since spherical harmonics of odd (resp. even) degrees are odd (resp. even) functions, it follows that $f$ is odd if and only if $\pi_m f = 0$ for each even $m$.

Specifically, the space $\mathscr{H}^1$ consists precisely of the restrictions to $\mathbb{S}^{n-1}$ of all linear functions on $\mathbb{R}^n$. The coordinate functions $u_1, \ldots, u_n$ (where $(u_1, \ldots, u_n) \in \mathbb{S}^{n-1}$) form an orthogonal basis of $\mathscr{H}^1$. Thus, for $f \in L^2(\mathbb{S}^{n-1})$, it has *no degree-one component*, i.e., its orthogonal projection $\pi_1 f = 0$, if and only if $f$ is orthogonal to each $u_i$, and if and only if

$$\int_{\mathbb{S}^{n-1}} u\, f(u)\, d\mathcal{H}^{n-1}(u) = 0.$$

3.2. **Isomorphism between spaces $C^\infty_{\text{odd}}(\mathbb{S}^{n-1})$.**

Write $C^\infty_{\text{odd}}(\mathbb{S}^{n-1})$ for the space of smooth odd functions $f : \mathbb{S}^{n-1} \to \mathbb{R}$, and define $C^\infty_{\text{even}}(\mathbb{S}^{n-1})$ analogously. Then $C^\infty(\mathbb{S}^{n-1})$ admits the direct sum decomposition

$$C^\infty(\mathbb{S}^{n-1}) = C^\infty_{\text{odd}}(\mathbb{S}^{n-1}) \oplus C^\infty_{\text{even}}(\mathbb{S}^{n-1}).$$

Recall that the *hemispherical transform* of $f \in C(\mathbb{S}^{n-1})$ is

$$\mathrm{T}\, f(u) = \int_{\mathbb{S}^{n-1}} \mathbb{1}_{\{\langle \theta, u\rangle \geq 0\}} f(\theta)\, d\mathcal{H}^{n-1}(\theta), \quad \forall\, u \in \mathbb{S}^{n-1}.$$

If $f$ is even, by the change of variables $\theta \mapsto -\theta$ and that $d\mathcal{H}^{n-1}(-\theta) = d\mathcal{H}^{n-1}(\theta)$, then

$$\mathrm{T}\, f(u) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} f(\theta)\, d\mathcal{H}^{n-1}(\theta), \quad \forall\, u \in \mathbb{S}^{n-1},$$

which is constant. What follows concentrate on the action of T on $C^\infty_{\text{odd}}(\mathbb{S}^{n-1})$.

**Lemma 3.2.** $\mathrm{T} : C^\infty_{\text{odd}}(\mathbb{S}^{n-1}) \longrightarrow C^\infty_{\text{odd}}(\mathbb{S}^{n-1})$ *is a linear isomorphism for each* $n \geq 3$*. Moreover,* $f \in C^\infty_{\text{odd}}(\mathbb{S}^{n-1})$ *has no degree-one spherical harmonic component if and only if* $\mathrm{T}\, f$ *does.*

*Proof.* In [19, (2.10) and (2.14)], Rubin showed that for every odd integer $m \geq 1$ and every spherical harmonic $Y_{m,k}$ of degree $m$, $\mathrm{T}\, Y_{m,k} = \lambda_m Y_{m,k}$, with eigenvalues

$$\lambda_m = \pi^{(n-2)/2} (-1)^{(m-1)/2} \frac{\Gamma(m/2)}{\Gamma((m+n)/2)}.$$

Since $\lambda_m \neq 0$ for all odd $m$, T is *injective* on $C^\infty_{\text{odd}}(\mathbb{S}^{n-1})$. Indeed, assume $f \in C^\infty_{\text{odd}}(\mathbb{S}^{n-1})$ and

$$f = \sum_{m \text{ odd}} \sum_{k=1}^{N(n,m)} c_{m,k} Y_{m,k}.$$

If $\mathrm{T}\, f = 0$, then

$$\sum_{m \text{ odd}} \sum_{k=1}^{N(n,m)} c_{m,k} \lambda_m Y_{m,k} = 0,$$

which forces $c_{m,k} = 0$ for all $m, k$ by the linear independence of $Y_{m,k}$. Hence, $f = 0$.

Next, we show that T is *surjective* on $C^\infty_{\text{odd}}(\mathbb{S}^{n-1})$. For $\forall g \in C^\infty_{\text{odd}}(\mathbb{S}^{n-1})$, write its spherical harmonic expansion (only odd degrees contribute) as

$$g = \sum_{m \text{ odd}} \sum_{k=1}^{N(n,m)} g_{m,k} Y_{m,k}.$$

Define

$$f := \sum_{m \text{ odd}} \sum_{k=1}^{N(n,m)} \lambda_m^{-1} g_{m,k} Y_{m,k},$$

where $\lambda_m \neq 0$ for all odd $m$ given as above. We show that $f \in C^\infty_{\text{odd}}(\mathbb{S}^{n-1})$. If so, then it follows immediately that $\mathrm{T}\, f = g$. Since only odd-degree spherical harmonics appear in its expansion, $f$ is odd. We aim to show that $f$ is smooth.

Recall that on $\mathbb{S}^{n-1}$ the Sobolev norm $H^s$ for $s \geq 0$ can be characterized via spherical harmonic coefficients (see Atkinson and Han [1, (3.98)]) as the following

$$\|g\|_{H^s}^2 = \sum_{m,k} (m + \frac{n-2}{2})^{2s} |g_{m,k}|^2.$$

Since $g$ is smooth, it lies in the Sobolev space $H^s(\mathbb{S}^{n-1})$. So the sum is finite for every $s \geq 0$.

Using Stirling's formula to the eigenvalues $\lambda_m$, we have the asymptotic estimate

$$|\lambda_m|^{-2} = O(m^n), \quad \text{as } m \to \infty,\ m \text{ odd},$$

and hence also

$$|\lambda_m|^{-2} = O((m + \frac{n-2}{2})^n), \quad \text{as } m \to \infty,\ m \text{ odd}.$$

So, for any $s \geq 0$,

$$\begin{aligned}
\|f\|_{H^s}^2 &= \sum_{m,k} (m + \frac{n-2}{2})^{2s} |\lambda_m|^{-2} |g_{m,k}|^2 \\
&= O\Big(\sum_{m,k} (m + \frac{n-2}{2})^{2s+n} |g_{m,k}|^2\Big) \\
&= O\big(\|g\|^2_{H^{s+n/2}(\mathbb{S}^{n-1})}\big) < +\infty.
\end{aligned}$$

Thus, $f \in H^s(\mathbb{S}^{n-1})$ for every $s \geq 0$. By the Sobolev embedding theorem on compact manifolds,

$$\bigcap_{s \geq 0} H^s(\mathbb{S}^{n-1}) = C^\infty(\mathbb{S}^{n-1}).$$

So $f$ is smooth, and therefore $f \in C^\infty_{\text{odd}}(\mathbb{S}^{n-1})$.

In light of that $\mathrm{T}\, Y_{m,k} = \lambda_m Y_{m,k}$, it follows that $\mathrm{T}$ is diagonalized by spherical harmonics. In particular, $\mathrm{T}$ acts on $\mathscr{H}^1$ as multiplication by $\lambda_1 \neq 0$. So, $\pi_1(\mathrm{T}\, f) = \mathrm{T}(\pi_1 f) = \lambda_1 \pi_1(f)$, which implies that $\mathrm{T}\, f$ has no degree-one component if and only if $f$ has no degree-one component. $\square$

### 3.3. $C^k$-boundedness.

In this part, we show that the hemispherical transform $\mathrm{T}$ is a bounded operator on the space $C^k(\mathbb{S}^{n-1})$. To this end, we first prove a general result on compact Lie groups.

**Lemma 3.3.** *Let $G$ be a compact Lie group and $\nu$ be a finite positive Borel measure on $G$. For each integer $k \geq 0$, define the operator $\mathcal{T}_\nu$ on $C^k(G)$ by*

$$(\mathcal{T}_\nu f)(\xi) = \int_G f(\xi\eta)\, d\nu(\eta), \quad \text{for } f \in C^k(G),\ \xi \in G.$$

*Then $\mathcal{T}_\nu : C^k(G) \to C^k(G)$ is bounded.*

*Proof.* Choose a basis $X_1, \dots, X_N$ of the Lie algebra of $G$, and let $X_1^R, \dots, X_N^R$ be the corresponding right-invariant vector fields. Since $f \in C^k(G)$ and the right-invariant vector fields act as differential operators of order $|I|$, the map

$$(\xi, \eta) \mapsto X_I^R f(\xi\eta)$$

is continuous on the compact space $G \times G$ for every multi-index $I$ with $|I| \leq k$. Since $\nu$ is a finite measure, the Lebesgue dominated convergence theorem gives that for all $\xi \in G$,

$$X_I^R(\mathcal{T}_\nu f)(\xi) = \int_G X_I^R f(\xi\eta)\, d\nu(\eta).$$

Since $G$ is compact and the integral is continuous in $\xi$, it follows that $X_I^R(\mathcal{T}_\nu f) \in C^0(G)$ for all $|I| \leq k$, which proves that $\mathcal{T}_\nu f \in C^k(G)$.

Since the right translation $\xi \mapsto \xi\eta$ is a diffeomorphism of $G$ onto itself, it follows that

$$\sup_{\xi \in G} |X_I^R f(\xi\eta)| = \sup_{g \in G} |X_I^R f(g)| = \|X_I^R f\|_{C^0(G)}.$$

Consequently,

$$\|X_I^R(\mathcal{T}_\nu f)\|_{C^0(G)} \leq \nu(G) \|X_I^R f\|_{C^0(G)}.$$

Finally, since $\{X_1^R, \dots, X_N^R\}$ forms a global frame of the tangent bundle, the standard $C^k-$norm on $G$ is equivalent to $\sum_{|I| \leq k} \|X_I^R(\cdot)\|_{C^0(G)}$. Summing the above estimate over all $|I| \leq k$ yields

$$\|\mathcal{T}_\nu f\|_{C^k(G)} \leq \nu(G) \|f\|_{C^k(G)}.$$

Thus, $\mathcal{T}_\nu : C^k(G) \to C^k(G)$ is bounded. ☐

**Lemma 3.4.** *For every integer $k \geq 0$, there exists a constant $C = C(n,k) > 0$ such that*

$$\|\mathrm{T} f\|_{C^k(\mathbb{S}^{n-1})} \leq C \|f\|_{C^k(\mathbb{S}^{n-1})}, \quad \textit{for each } f \in C^k(\mathbb{S}^{n-1}).$$

*Proof.* Let $G = \mathrm{SO}(n)$ and let $\pi : G \to \mathbb{S}^{n-1}$ be given by $\pi(\xi) = \xi e_n$, where $e_n = (0, 0, \dots, 1)$. For $f \in C^k(\mathbb{S}^{n-1})$, define its pullback $\widetilde{f} = f \circ \pi \in C^k(\mathbb{S}^{n-1})$. Let $E = \{\theta \in \mathbb{S}^{n-1} : \langle \theta, e_n \rangle \geq 0\}$.

First, we construct a finite positive Borel measure $\nu$ on $G$ such that its pushforward satisfies $\pi_* \nu = \mathcal{H}^{n-1}|_E$. Indeed, let $m_G$ be the Haar probability measure on $G$. Since $\pi(\xi) = \xi e_n$ identifies $\mathbb{S}^{n-1}$ with $G/\mathrm{SO}(n-1)$, the pushforward $\pi_* m_G$ is precisely the normalized rotation-invariant measure on $\mathbb{S}^{n-1}$, that is,

$$\pi_* m_G = \frac{1}{\mathcal{H}^{n-1}(\mathbb{S}^{n-1})} \mathcal{H}^{n-1}.$$

For any Borel set $B \subset G$, define

$$\nu(B) = \mathcal{H}^{n-1}(\mathbb{S}^{n-1})\, m_G(B \cap \pi^{-1}(E)).$$

Then $\nu(G) = \frac{1}{2}\mathcal{H}^{n-1}(\mathbb{S}^{n-1}) < \infty$, and for any Borel set $A \subset \mathbb{S}^{n-1}$,

$$\pi_* \nu(A) = \mathcal{H}^{n-1}(\mathbb{S}^{n-1}) m_G(\pi^{-1}(A \cap E)) = \mathcal{H}^{n-1}(A \cap E).$$

Thus, $\pi_* \nu = \mathcal{H}^{n-1}|_E$.

From the $\mathrm{SO}(n)-$invariance of the spherical measure and that $\pi_* \nu = \mathcal{H}^{n-1}|_E$, for any $\xi \in G$ we have

$$\widetilde{\mathrm{T}\,f}(\xi) = \mathrm{T}\,f(\xi e_n) = \int_{\xi E} f(\theta)\,d\mathcal{H}^{n-1}(\theta) = \int_E f(\xi\theta)\,d\mathcal{H}^{n-1}(\theta)$$
$$= \int_G f(\xi\eta e_n)\,d\nu(\eta) = \int_G \widetilde{f}(\xi\eta)\,d\nu(\eta) = \mathcal{T}_\nu \widetilde{f}(\xi).$$

Since $\pi : G \to \mathbb{S}^{n-1}$ is a smooth submersion with compact fibre $\mathrm{SO}(n-1)$, there exists $\delta > 0$ such that for every $f \in C^k(G)$

$$\delta \|\widetilde{f}\|_{C^k(G)} \le \|f\|_{C^k(\mathbb{S}^{n-1})} \le \delta^{-1}\|\widetilde{f}\|_{C^k(G)}.$$

By Lemma 3.3, the operator $\mathcal{T}_\nu$ is bounded on $C^k(G)$. Thus,

$$\|\mathrm{T}\,f\|_{C^k(\mathbb{S}^{n-1})} \le \delta^{-1}\|\widetilde{\mathrm{T}\,f}\|_{C^k(G)} = \delta^{-1}\|\mathcal{T}_\nu\widetilde{f}\|_{C^k(G)} \le \delta^{-1}C'\|\widetilde{f}\|_{C^k(G)} \le \delta^{-2}C'\|f\|_{C^k(\mathbb{S}^{n-1})}.$$

Let $C = \delta^{-2}C'$. This completes the proof. □

## 4. **The stability of Morse functions**

To prove of the main result, we use the existence of our desired Morse functions. For basic facts on Morse theory, please refer to, e.g., Milnor [16].

Let $M$ be a smooth manifold and $f \in C^2(M)$. If $df(p) = 0$, then $p$ is called a *critical point* of $f$. Let $(x^i)$ be local coordinates around $p$. The *Hessian* of $f$ at $p$, denoted by $\mathrm{Hess}_M f(p)$, is the symmetric bilinear form on $T_pM$ whose matrix with respect to the coordinate basis $(\frac{\partial}{\partial x^i}|_p)$ is $(\frac{\partial^2 f}{\partial x^i \partial x^j}(p))$. The critical point $p$ is called *nondegenerate*, if the matrix $(\frac{\partial^2 f}{\partial x^i \partial x^j}(p))$ is invertible.

**Definition 4.1.** Let $M$ be a smooth manifold and $f \in C^2(M)$. The function $f$ is called a *Morse function*, if every critical point of $f$ is nondegenerate.

The property of being Morse is invariant under a diffeomorphism: Given a diffeomorphism $\Phi : N \to M$, a function $f : M \to \mathbb{R}$ is Morse if and only if $f \circ \Phi : N \to \mathbb{R}$ is Morse. Moreover, if $f$ has critical points $\{p_i\}$, then $f \circ \Phi$ has critical points $\{\Phi^{-1}(p_i)\}$.

A crucial property of Morse functions which we use is their stability under sufficiently small perturbations in the $C^2-$topology. We first prove this stability in a local setting.

**Lemma 4.2.** *Let $U \subseteq \mathbb{R}^n$ be open, $f \in C^2(U)$, and let $p \in U$ be a nondegenerate critical point of $f$. Then there exists a real $\delta > 0$ and an open ball $V$ with $p \in V$ and its closure $\bar{V} \subset U$, such that every $g \in C^2(U)$ with $\|g - f\|_{C^2(\bar{V})} < \delta$ has exactly one critical point in $V$ wihch is also nondegenerate.*

*Proof.* Translating the coordinates if necessary, we assume that $p = 0$. Choose a sufficiently small open ball $V$ centered at $0$ with $\bar{V} \subset U$, such that $\nabla f$ does not vanish on $\partial V$ and $\nabla^2 f$ is nondegenerate on $\bar{V}$.

**Step 1.** Show that there exists a point $x \in V$ such that $\nabla g(x) = 0$.

By the compactness of $\bar{V}$, we choose $\delta > 0$ sufficiently small such that for every $g \in C^2(U)$ with $\|g - f\|_{C^2(\bar{V})} < \delta$ and for all $t \in [0, 1]$, (i). $\nabla f + t\nabla(g-f) \neq 0$ on $\partial V$,

(ii). $\nabla^2 f + t\nabla^2(g-f)$ is nondegenerate on $\bar{V}$.

Let

$$F(t,x) = \nabla f(x) + t\nabla(g-f)(x), \quad \text{for } (t,x) \in [0,1]\times V.$$

We aim to show that $F(1,x)=0$ for some $x \in V$. Let

$$S = \{t \in [0,1] : \text{ there exists } x \in V \text{ such that } F(t,x)=0\}.$$

It suffices to show $S=[0,1]$. Clearly $0 \in S$ because $F(0,0)=0$.

We first show that $S$ is open. Suppose $t_0 \in S$, and let $x_0 \in V$ be such that $F(t_0,x_0)=0$. Then

$$D_y F(t_0,x_0) = \nabla^2 f(x_0) + t_0\nabla^2(g-f)(x_0)$$

is nondegenerate by the choice of $\delta$. Therefore, by the implicit function theorem, there is a neighborhood $\mathcal{N}(t_0)$ of $t_0$ in $[0,1]$ such that for every $t \in \mathcal{N}(t_0)$, there exists $x \in V$ that solves $F(t,x)=0$. That is, $\mathcal{N}(t_0) \subseteq S$. Hence $S$ is open in $[0,1]$.

We next show that $S$ is closed. Let $t_j \in S$ and $t_j \to t$. Then by a routine compactness argument, there exists $x \in \bar{V}$ with $F(t,x)=0$. By our choice of $\delta$, $x \notin \partial V$, hence $x \in V$. This shows $S$ is closed in $[0,1]$.

Hence $S$ is nonempty, open and closed, and therefore $S=[0,1]$. In particular, $1 \in S$ as desired. Moreover, by (ii), the Hessian of $g$ is nondegenerate at any critical point of $g$ in $V$.

**Step 2.** Show that there is exactly one $x \in V$ with $\nabla g(x)=0$ for sufficiently small $\delta$.

We argue by contradiction. Assume there is a sequence $\{g_m\}_m$ such that $\|g_m - f\|_{C^2(\bar{V})} \leq \frac{1}{m}$ and that $\nabla g_m(x_m) = \nabla g_m(y_m) = 0$ for some $x_m \neq y_m \in V$. Then we have $\lim_{m\to\infty} x_m = 0$. Indeed, if not, then a subsequence of $\{x_m\}$ converges to $x \in \bar{V}\backslash\{0\}$ with $\nabla f(x)=0$, contradicting the assumption. Similarly, $\lim_{m\to\infty} y_m = 0$.

Set

$$u_m := \frac{x_m - y_m}{\|x_m - y_m\|}.$$

Since $\bar{V}$ is compact, $\nabla^2 f$ is uniformly continuous on $\bar{V}$; Since $x_m, y_m \to 0$, we have

$$\sup_{0\leq t\leq 1}\left\|\nabla^2 f(y_m + t(x_m-y_m)) - \nabla^2 f(0)\right\| \leq \omega_{\nabla^2 f}(\max(\|x_m\|,\|y_m\|)) \to 0, \tag{4.1}$$

where $\omega_{\nabla^2 f}$ is the modulus of continuity of $\nabla^2 f$ on $\bar{V}$. So we have,

$$\begin{aligned}\frac{\nabla g_m(x_m) - \nabla g_m(y_m)}{\|x_m - y_m\|} &= \int_0^1 \nabla^2 g_m(y_m + t(x_m-y_m))u_m\,dt \\ &= \int_0^1 \nabla^2 f(y_m + t(x_m-y_m))u_m\,dt + O(\frac{1}{m}) \\ &= \nabla^2 f(0)u_m + o(1),\end{aligned}$$

where we used $\|g_m - f\|_{C^2(\bar{V})} \leq \frac{1}{m}$ for the second equality and (4.1) for the last equality. However,

$$\|\nabla^2 f(0)u_m\| \geq \|(\nabla^2 f(0))^{-1}\|^{-1},$$

where $\|\cdot\|$ is the operator norm. Hence

$$0 = \liminf_{m\to\infty} \left\| \frac{\nabla g_m(x_m) - \nabla g_m(y_m)}{\|x_m - y_m\|} \right\| \geq \|(\nabla^2 f(0))^{-1}\|^{-1} > 0,$$

a contradiction. □

The stability for Morse functions now follows from the previous local result.

**Lemma 4.3.** *Let $M$ be a compact smooth manifold without boundary and $f \in C^2(M)$ be Morse. Then there exists $\delta > 0$ such that every $g \in C^2(M)$ with $\|g - f\|_{C^2(M)} < \delta$ is also Morse and has the same number of critical points as $f$.*

*Proof.* Let $p_1, \dots, p_\ell$ be the critical points of $f$ with $\ell \geq 1$. Choose pairwise disjoint coordinate neighborhoods $Q_j$ of the points $p_j$, which are small enough such that $p_j$ is the only critical point of $f$ in $Q_j$. Fix coordinates $\phi_j : Q_j \to \mathbb{R}^{\dim M}$ with $\phi_j(p_j) = 0$, and shrink $Q_j$ so that the Hessian of $f \circ \phi_j^{-1}$ is nondegenerate throughout $\phi_j(Q_j)$. Applying Lemma 4.2 to $f \circ \phi_j^{-1}$ gives an open ball $V_j$ with its closure $\bar{V}_j \subset \phi_j(Q_j)$ such that every sufficiently small $C^2$ perturbation has exactly one non-degenerate critical point in $V_j$. Thus, after choosing one common perturbation size $\delta$, $g$ has exactly one non-degenerate critical point in each

$$U_j := \phi_j^{-1}(V_j).$$

In light of $f$ has no critical point in the compact set $M \setminus \bigcup_{j=1}^{\ell} U_j$, by taking $\delta$ small enough, $g$ also has no critical point in $M \setminus \bigcup_{j=1}^{\ell} U_j$. Hence, $g$ has exactly $\ell$ critical points, and they are non-degenerate, as desired. □

## 5. Proof of main results

We prove Theorem 1.1 in this section. Lemma 5.1 shows that the existence of certain Morse functions on the sphere gives the desired convex body. Lemma 5.3 then constructs such functions explicitly.

**Lemma 5.1.** *Let $n \geq 3$. Suppose that $f \in C^\infty_{\mathrm{odd}}(\mathbb{S}^{n-1})$ is Morse with exactly two critical points $p$ and $-p$, and that*

$$\int_{\mathbb{S}^{n-1}} u f(u)\, d\mathcal{H}^{n-1}(u) = 0. \tag{5.1}$$

*Then for all sufficiently small $\varepsilon > 0$,*

$$K_\varepsilon := \{ru : u \in \mathbb{S}^{n-1},\ 0 \leq r \leq (1 + \varepsilon(\mathrm{T}^{-1} f)(u))^{1/(n+1)}\}$$

*is a smooth strictly convex body with centroid $0$ and there exists exactly one hyperplane section of $K_\varepsilon$ with centroid $0$.*

*Proof.* We divide the proof into two steps.

**Step 1.** Show that $K_\varepsilon$ is a smooth strictly convex body with centroid $0$.

For brevity, set $h = \mathrm{T}^{-1} f$. By Lemma 3.2, $h$ is odd and smooth. For sufficiently small $|\varepsilon| > 0$, we have $1 + \varepsilon h > 0$. Lemma 2.4 gives

$$(1 + \varepsilon h)^{1/(n+1)} = 1 + \frac{\varepsilon}{n+1} h + O(\varepsilon^2) \quad \text{in } C^2(\mathbb{S}^{n-1}).$$

Thus, $(1 + \varepsilon h)^{1/(n+1)} \to 1$ in $C^2(\mathbb{S}^{n-1})$ as $\varepsilon \to 0$, and $K_\varepsilon$ is a smooth strictly convex body with radial function $\rho_{K_\varepsilon}(u) = (1 + \varepsilon h(u))^{1/(n+1)}$ by Lemma 2.3.

Since $f$ has no degree-one component by condition (5.1) and $\mathrm{T}\, h = f$, by Lemma 3.2, it yields that $h$ has no degree-one component, i.e.,

$$\int_{\mathbb{S}^{n-1}} u h(u) \, d\mathcal{H}^{n-1}(u) = 0.$$

So

$$\begin{aligned}
\int_{K_\varepsilon} x \, d\mathcal{H}^n(x) &= \int_{\mathbb{S}^{n-1}} \Big( \int_0^{\rho_{K_\varepsilon}(u)} u \, t^n \, dt \Big) \, d\mathcal{H}^{n-1}(u) \\
&= \frac{1}{n+1} \int_{\mathbb{S}^{n-1}} u \rho_{K_\varepsilon}(u)^{n+1} \, d\mathcal{H}^{n-1}(u) \\
&= \frac{1}{n+1} \int_{\mathbb{S}^{n-1}} u (1 + \varepsilon h(u)) \, d\mathcal{H}^{n-1}(u) = 0.
\end{aligned}$$

That is, $c(K_\varepsilon) = 0$.

**Step 2.** Show that $K_\varepsilon$ has exactly one hyperplane section of $K_\varepsilon$ with centroid 0.

By formula (3.1) and Lemma 2.4, the half-space volume function

$$\begin{aligned}
A_{K_\varepsilon} &= \frac{1}{n} \mathrm{T}\big( (1 + \varepsilon h)^{\frac{n}{n+1}} \big) \\
&= \frac{1}{n} \mathrm{T} \big( 1 + \frac{n}{n+1} \varepsilon h + R_\varepsilon \big) \\
&= \frac{\mathcal{H}^{n-1}(\mathbb{S}^{n-1})}{2n} + \frac{\varepsilon}{n+1} f + \frac{1}{n} \mathrm{T}(R_\varepsilon),
\end{aligned} \tag{5.2}$$

where $\|R_\varepsilon\|_{C^2(\mathbb{S}^{n-1})} = O(\varepsilon^2)$.

Let

$$g_\varepsilon = \frac{n+1}{\varepsilon} \big( A_{K_\varepsilon} - \frac{\mathcal{H}^{n-1}(\mathbb{S}^{n-1})}{2n} \big).$$

Using (5.2) and Lemma 3.4, we have

$$\|g_\varepsilon - f\|_{C^2(\mathbb{S}^{n-1})} = \frac{n+1}{n\varepsilon} \| \mathrm{T}(R_\varepsilon) \|_{C^2(\mathbb{S}^{n-1})} = O(\varepsilon).$$

Lemma 4.3 therefore implies that $g_\varepsilon$, and hence $A_{K_\varepsilon}$, is Morse with exactly two critical points for all sufficiently small $\varepsilon > 0$.

Since $A_K(-u) = \mathcal{H}^n(K) - A_K(u)$, these two critical points are antipodal. Denote them by $u$ and $-u$. The directions $u$ and $-u$ determine the same hyperplane $u^\perp$, and Lemma 2.1 shows that $K_\varepsilon \cap u^\perp$ has centroid 0. Since $A_{K_\varepsilon}$ has exactly two critical points, any other hyperplane section of $K_\varepsilon$ has centroid not at the origin. This completes the proof. $\square$

It remains to show that a Morse function $f$ involved in Lemma 5.1 exists. For this aim, we first prove the following lemma.

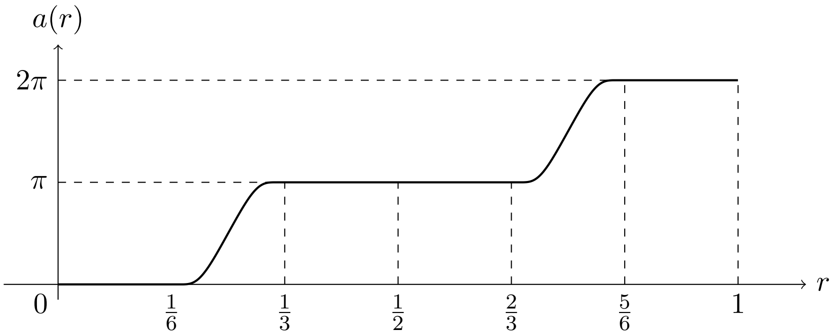


FIGURE 1. Graph of the function $a$.

**Lemma 5.2.** *There exists a smooth even function $\alpha : [-1,1] \to \mathbb{R}$ such that*

$$\int_{-1}^{1} (1-t^2)^{(n-1)/2} e^{i\alpha(t)}\, dt = 0. \tag{5.3}$$

*Proof.* Let $w(t) = (1-t^2)^{(n-1)/2}$, $W = \int_0^1 w(t)\, dt$, and

$$s(t) = W^{-1} \int_0^t w(\tau)\, d\tau, \quad 0 \le t \le 1.$$

Then $s$ is smooth in $(0,1)$ and increasing on $[0,1]$, with $s(0) = 0$ and $s(1) = 1$.

Let $a \in C^\infty([0,1])$ be as in Figure 1. In particular, we require that $a = 0$ near 0; $a = \pi$ near $\frac{1}{2}$; $a = 2\pi$ near 1; and

$$a(x + \frac{1}{2}) = \pi + a(x) \text{ for all } x \in [0, \frac{1}{2}] \tag{5.4}$$

Define $\alpha(t) = a(s(t))$ for $0 \le t \le 1$, and extend it evenly to $[-1,1]$. Since $a$ is constant near 0 and 1, the extension is globally smooth. Using $w(t)\, dt = W\, ds(t)$ and (5.4), we have

$$\int_{-1}^{1} w(t) e^{i\alpha(t)}\, dt = 2W \int_0^1 e^{ia(r)}\, dr$$
$$= 2W \int_0^{\frac{1}{2}} e^{ia(r)}\, dr + 2W \int_0^{\frac{1}{2}} e^{i(\pi + a(r))}\, dr = 0.$$

This proves the lemma. $\square$

Now we construct the desired $f$. We start with an odd linear function which is Morse with exactly two nondegenerate critical points, while it has a nonzero degree-one component. Then we compose it with a suitably odd diffeomorphism of the sphere, which preserves the Morse property and the number of critical points while arranging for the first moment to vanish.

Fix $n \ge 3$. Write each point $u$ of $\mathbb{S}^{n-1} \subset \mathbb{R}^n$ as

$$u = (x_1, x_2, y, t) \in \mathbb{R} \oplus \mathbb{R} \oplus \mathbb{R}^{n-3} \oplus \mathbb{R}, \qquad x_1^2 + x_2^2 + |y|^2 + t^2 = 1.$$

We look $t$ as the altitude of the point $u$. The construction rotates the $(x_1, x_2)$-plane by an angle $\alpha(t)$ determined by Lemma 5.2.

**Lemma 5.3.** *Let $n \geq 3$. There exists a Morse function $f \in C^\infty_{\mathrm{odd}}(\mathbb{S}^{n-1})$ which has exactly two critical points $p$ and $-p$, and satisfies*

$$\int_{\mathbb{S}^{n-1}} uf(u)\, d\mathcal{H}^{n-1}(u) = 0.$$

*Proof.* We divide the proof into two steps.

**Step 1.** Construct a Morse function $f$.

Let $\alpha$ be as in Lemma 5.2. For $\theta \in \mathbb{R}$, write

$$\phi_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Define

$$\Phi : \mathbb{S}^{n-1} \to \mathbb{S}^{n-1}, \quad \Phi(x_1, x_2, y, t) = (\phi_{\alpha(t)}(x_1, x_2), y, t).$$

Then $\Phi$ is a smooth diffeomorphism, with inverse obtained by replacing $\alpha(t)$ by $-\alpha(t)$. Since $\alpha$ is even, $\Phi$ is odd. Let $f$ be the first coordinate of $\Phi$, i.e.,

$$f(x_1, x_2, y, t) = x_1 \cos\alpha(t) - x_2 \sin\alpha(t), \quad (x_1, x_2, y, t) \in \mathbb{S}^{n-1}.$$

**Step 2.** Show that $f$ is the desired function.

By the construction, $f = x_1 \circ \Phi$. Since the function $x_1$ is Morse with exactly two critical points ($\pm e_1 = (\pm 1, 0, \ldots, 0)$) and $\Phi$ is a diffeomorphism, $f = x_1 \circ \Phi$ is Morse with precisely the two critical points $\Phi^{-1}(e_1)$ and $\Phi^{-1}(-e_1)$. Also, $f$ is odd and smooth because both $x_1$ and $\Phi$ are odd and smooth.

It remains to prove the first moment of $f$ vanish. For

$$u = (x_1, x_2, y, t) \in \mathbb{S}^{n-1} \setminus \{\pm e_n\},$$

write

$$v = (v_1, \ldots, v_{n-1}) = \frac{(x_1, x_2, y)}{\sqrt{1-t^2}} \in \mathbb{S}^{n-2}.$$

Then $u_i = \sqrt{1-t^2}\, v_i$ for $1 \leq i \leq n-1$, $u_n = t$, and

$$f(u) = \sqrt{1-t^2}\,(v_1 \cos\alpha(t) - v_2 \sin\alpha(t)).$$

So

$$\int_{\mathbb{S}^{n-1}} u_n f(u)\, d\mathcal{H}^{n-1}(u) = \int_{\mathbb{S}^{n-2}} \int_{-1}^{1} t(v_1 \cos\alpha(t) - v_2 \sin\alpha(t))(1-t^2)^{\frac{n-2}{2}}\, dt\, d\mathcal{H}^{n-2}(v) = 0,$$

where the last equality holds because $\alpha$ is even; for $1 \leq i \leq n-1$,

$$\begin{aligned} &\int_{\mathbb{S}^{n-1}} u_i f(u)\, d\mathcal{H}^{n-1}(u) \\ =&\int_{\mathbb{S}^{n-2}} \int_{-1}^{1} v_i(v_1 \cos\alpha(t) - v_2 \sin\alpha(t))(1-t^2)^{(n-1)/2}\, dt\, d\mathcal{H}^{n-2}(v) \\ =&2\int_{\mathbb{S}^{n-2}} v_i \int_{0}^{1} (v_1 \cos\alpha(t) - v_2 \sin\alpha(t))(1-t^2)^{(n-1)/2}\, dt\, d\mathcal{H}^{n-2}(v) = 0, \end{aligned}$$

where the last equality follows from the real and imaginary parts of (5.3). Therefore,

$$\int_{\mathbb{S}^{n-1}} uf(u)\, d\mathcal{H}^{n-1}(u) = 0$$

and the lemma follows. $\square$

*Proof of Theorem 1.1.* For every $n \geq 3$, choose $f$ as in Lemma 5.3. Then Lemma 5.1 gives the desired convex bodies. Therefore, $\mu(n) \leq 1$ for $n \geq 3$..

By Lemma 2.2, $\mu(n) \geq 1$ for all $n \geq 2$. Therefore, $\mu(n) = 1$ for all $n \geq 3$. Lemma 2.5 gives $\mu(2) \geq 3$, and Lemma 2.6 gives $\mu(2) \leq 3$, hence $\mu(2) = 3$. $\square$

**Conflict of Interest**: We declare that we have no conflict of interest.

**Data Availability**: Not applicable.

## References


[1] K. Atkinson, W. Han, Spherical harmonics and approximations on the unit sphere: an introduction, Lecture Notes in Mathematics, 2044, Springer, Heidelberg, 2012. 9

[2] R. Bose, A note on the convex oval, Bull. Calcutta Math. Soc. 27 (1935) 55–60. 1, 5

[3] F. Dai, Y. Xu, Approximation theory and harmonic analysis on spheres and balls, Springer Monographs in Mathematics, Springer, New York, 2013. 7

[4] E. Ehrhart, Une généralisation du théorème de Minkowski, C. R. Acad. Sci. Paris 240 (1955) 483–485. 1, 5

[5] W. Fenchel (ed.), Convexity: proceedings of the colloquium on convexity, Copenhagen, 1965. 1

[6] P. Funk, Über eine geometrische anwendung der abelschen integralgleichung, Math. Ann. 77 (1915) 129–135. 6

[7] R. Gardner, A positive answer to the Busemann–Petty problem in three dimensions, Ann. of Math. 140 (1994) 435–447. 2

[8] H. Groemer, Geometric applications of Fourier series and spherical harmonics, Cambridge University Press, Cambridge, 1996. 7

[9] B. Grünbaum, On some properties of convex sets, Colloq. Math. 8 (1961) 39–42. 1, 3, 4

[10] B. Grünbaum, Measures of symmetry for convex sets, Proc. Sympos. Pure Math., Vol. VII, Amer. Math. Soc., 1963, pp. 233–270. 3

[11] B. Guan, J. Spruck, Boundary-value problems on $\mathbb{S}^n$ for surfaces of constant Gauss curvature, Ann. of Math. 138 (1993) 601–624. 4

[12] J. Haddad, C. Jiménez, R. Villa, Centroids of sections of convex bodies and Lusternik–Schnirelmann category, arXiv:2511.22393, 2025. 3

[13] Y. Huang, S. Myroshnychenko, K. Tatarko, V. Yaskin, On hyperbolic and functional analogues of questions of Grünbaum and Loewner, arXiv:2606.04482, 2026. 2

[14] A. Koldobsky, Fourier analysis in convex geometry, Amer. Math. Soc., 2005. 4

[15] M. Meyer, S. Reisner, Characterizations of ellipsoids by section-centroid location, Geom. Dedicata 31 (1989) 345–355. 4

[16] J. Milnor, Morse Theory, Annals of Mathematics Studies, vol. 51, Princeton University Press, 1963. 11

[17] S. Myroshnychenko, K. Tatarko, V. Yaskin, Answers to questions of Grünbaum and Loewner, Adv. Math. 461 (2025) Paper No. 110081. 2, 4, 5

[18] Z. Patáková, M. Tancer, U. Wagner, Barycentric cuts through a convex body, Discrete Comput. Geom. 68 (2022) 1133–1154. 3

[19] B. Rubin, Inversion and characterization of the hemispherical transform, J. Anal. Math. 77 (1999) 105–128. 6, 8

[20] R. Schneider, Convex bodies: the Brunn–Minkowski theory, Cambridge University Press, 2014. 7

[21] H. Steinhaus, Quelques applications des principes topologiques à la géométrie des corps convexes, Fund. Math. 41 (1955) 284–290. 4

[22] V. Yaskin, The Busemann–Petty problem in hyperbolic and spherical spaces, Adv. Math. 203 (2006) 537–553. 2

[23] G. Zhang, A positive solution to the Busemann–Petty problem in $\mathbb{R}^4$, Ann. of Math. 149 (1999) 535–543. 2